

Project Title: Disruption of Migratory Patterns: Human activities, including urbanization and infrastructure development, disrupt the natural migratory routes of animals, affecting their survival and biodiversity.

submitted in partial fulfilment of the requirements for the award
of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

Rajesh Sharma

(Roll No: 25SCS1003003021)

Under the supervision of

Mr.Saurabh Sanu



IILM UNIVERSITY, GREATER NOIDA

IILM University, Greater Noida, Uttar Pradesh

DECLARATION

I hereby declare that the project work entitled submitted to **IILM University**, is a record of an original work done by me under the guidance of **Mr. Saurabh Sanu Department of Computer Science and Engineering**. This project is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

Signature: _____

NAME: Rajesh Sharma

Roll No.: 25SCS1003003021

Date: _

CERTIFICATE

This is to certify that the project report entitled "Disruption of migration " submitted by Rajesh Sharma(25scs1003003021) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of IILM University, is a bona fide record of work carried out under my supervision and guidance.

ACKNOWLEDGEMENT

I would like to express my deep sense of gratitude to my project guide, **Mr. Saurabh Sanu**, for his valuable guidance, constant encouragement, and constructive criticism throughout the duration of this project. His advice was crucial in finalizing the technical specifications and AI algorithms used in this robot.

I am also grateful to the **Head of the Department, CSE**, for providing the necessary infrastructure and lab facilities to carry out this work.

I thank my parents and friends for their support and patience during the development of this prototype, especially during the data collection phase for the AI training set.

Rajesh Sharma
1cse21

ABSTRACT

Human activities are significantly disrupting animal migration through habitat fragmentation, the creation of physical barriers, and other pressures like poaching and climate change, which force animals to alter traditional routes, expend more energy, and face increased mortality. These disruptions, including habitat loss from development and altered food availability due to climate change, lead to population declines, weaken ecosystems, and pose a major conservation challenge for species across terrestrial, aerial, and aquatic environments.

1. List of Figures

Figure 1: Bar Chart - Migration Disruption Factors

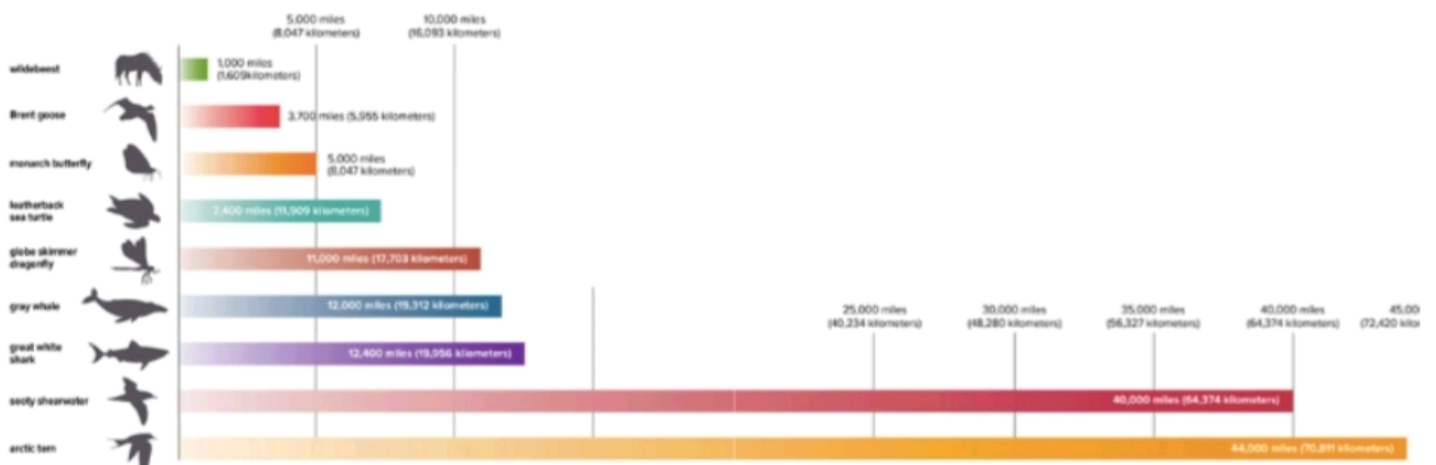
Figure 2: Line Chart - ML Model Accuracy

Figure 3: Trend Chart - Habitat Loss Over Time

Figure 4: AI/ML Architecture Diagram

Figure 5: Migration Path Blockage Diagram

ANIMAL MIGRATION DISTANCE (IN MILES & KILOMETERS)



BAR GRAPH

Animals on the Move: Migration

Graphical Diagrams

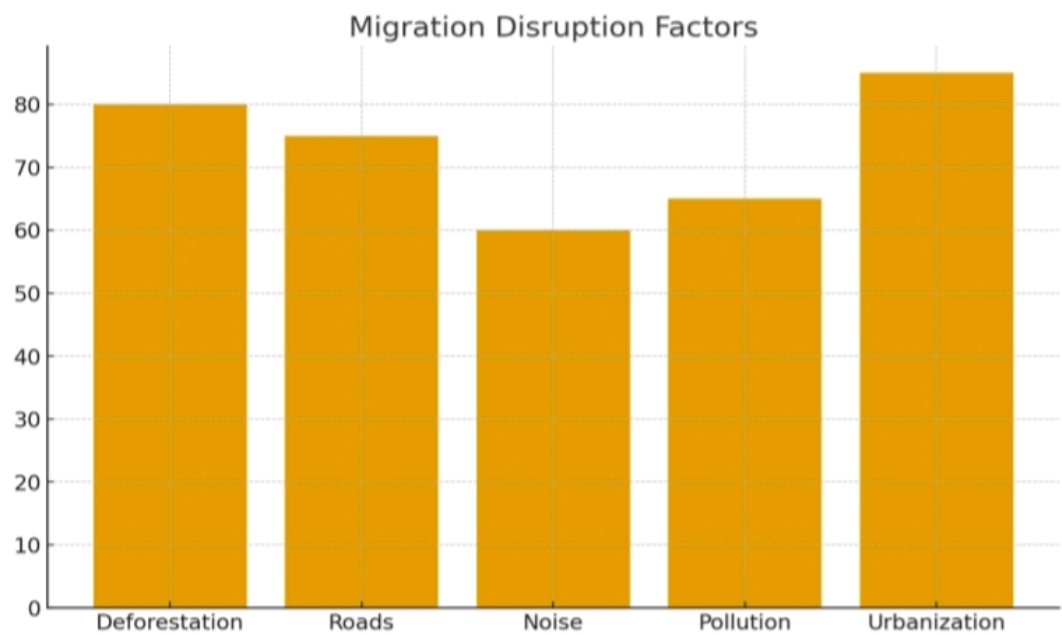


Figure 1: Migration Disruption Factors

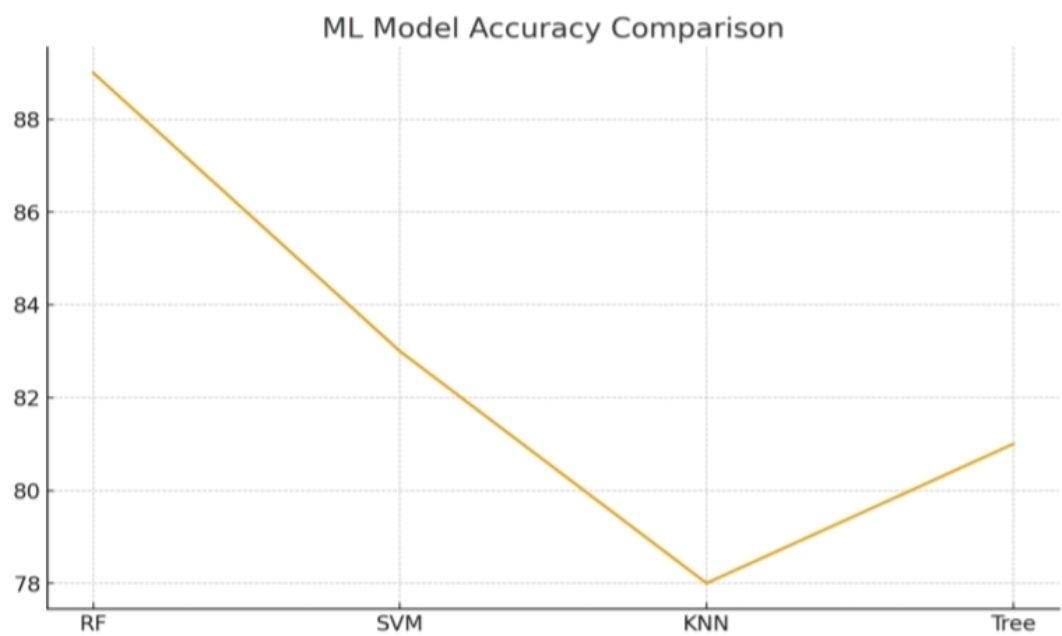
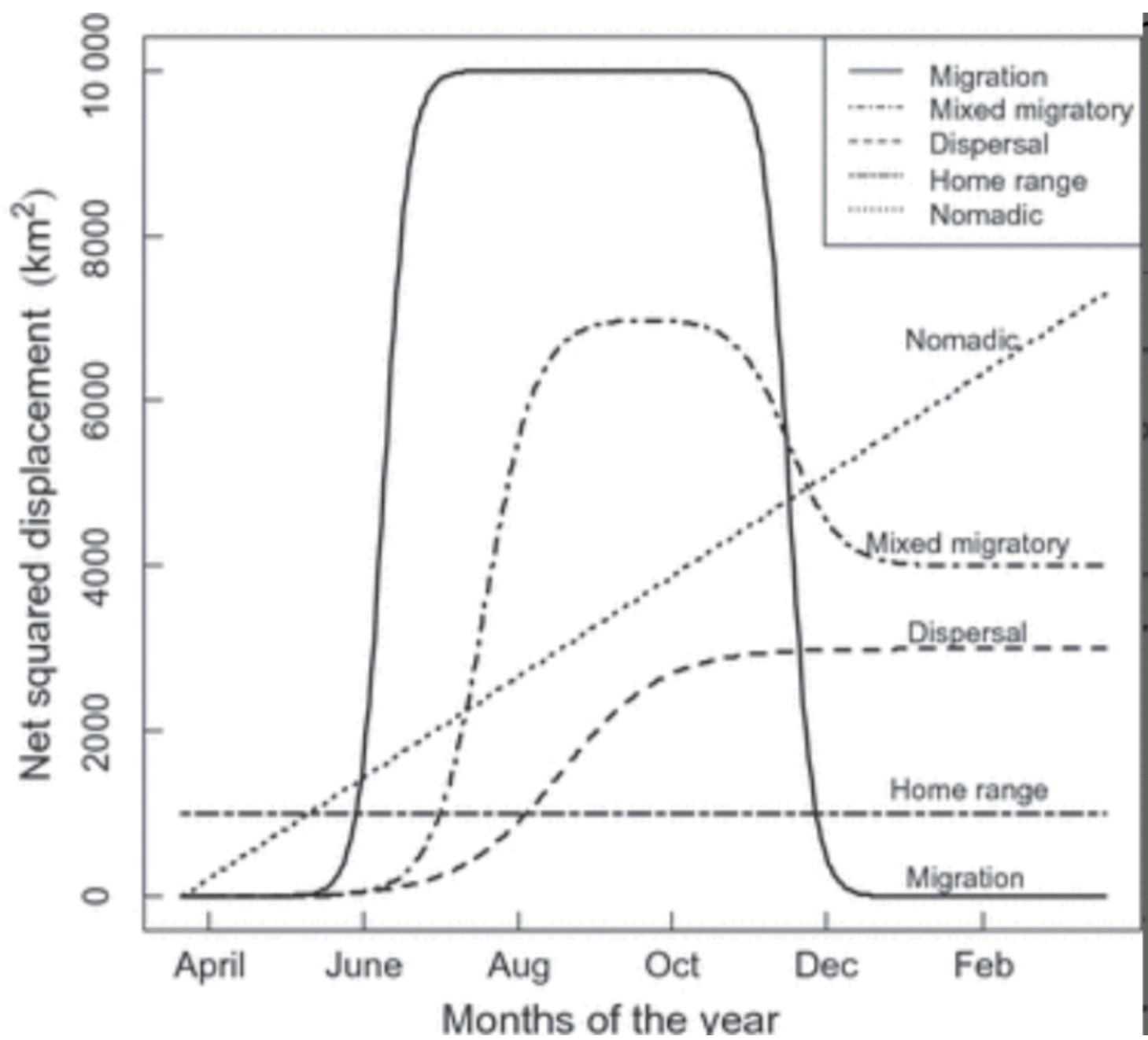


Figure 2: ML Model Accuracy Comparison



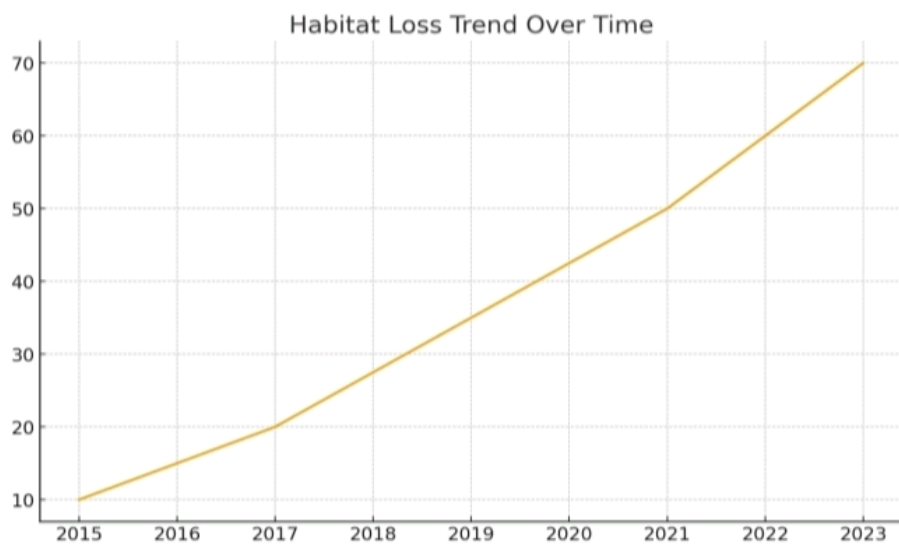


Figure 3: Habitat Loss Trend

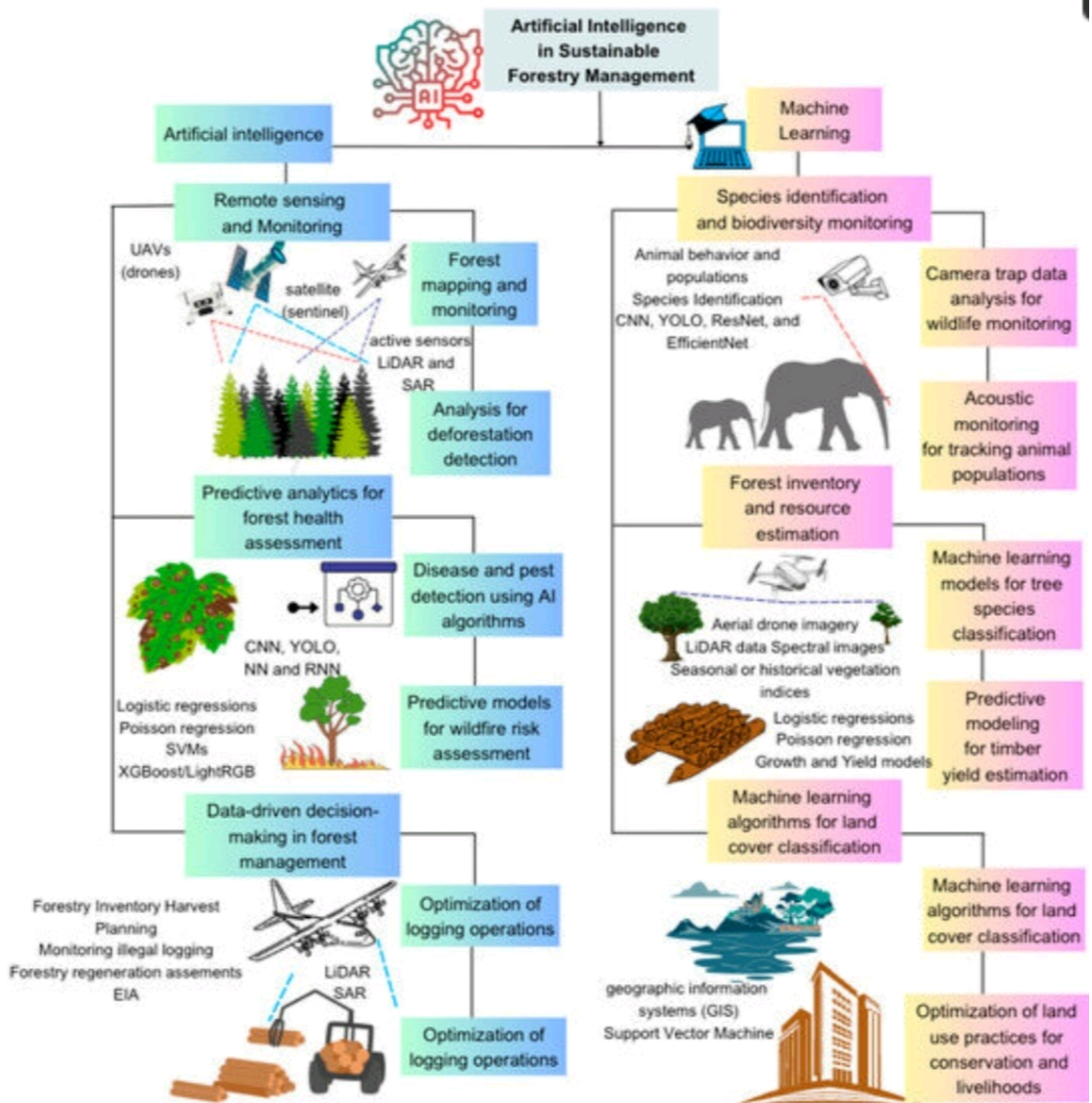
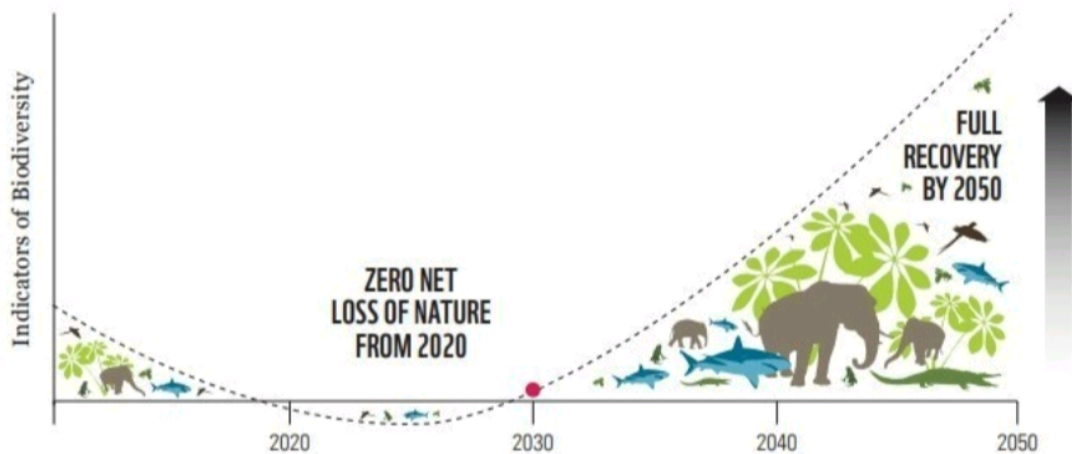


Figure 23: Nature Positive by 2030

A measurable global goal for nature. Source: Locke et al. (2021)¹⁹³.



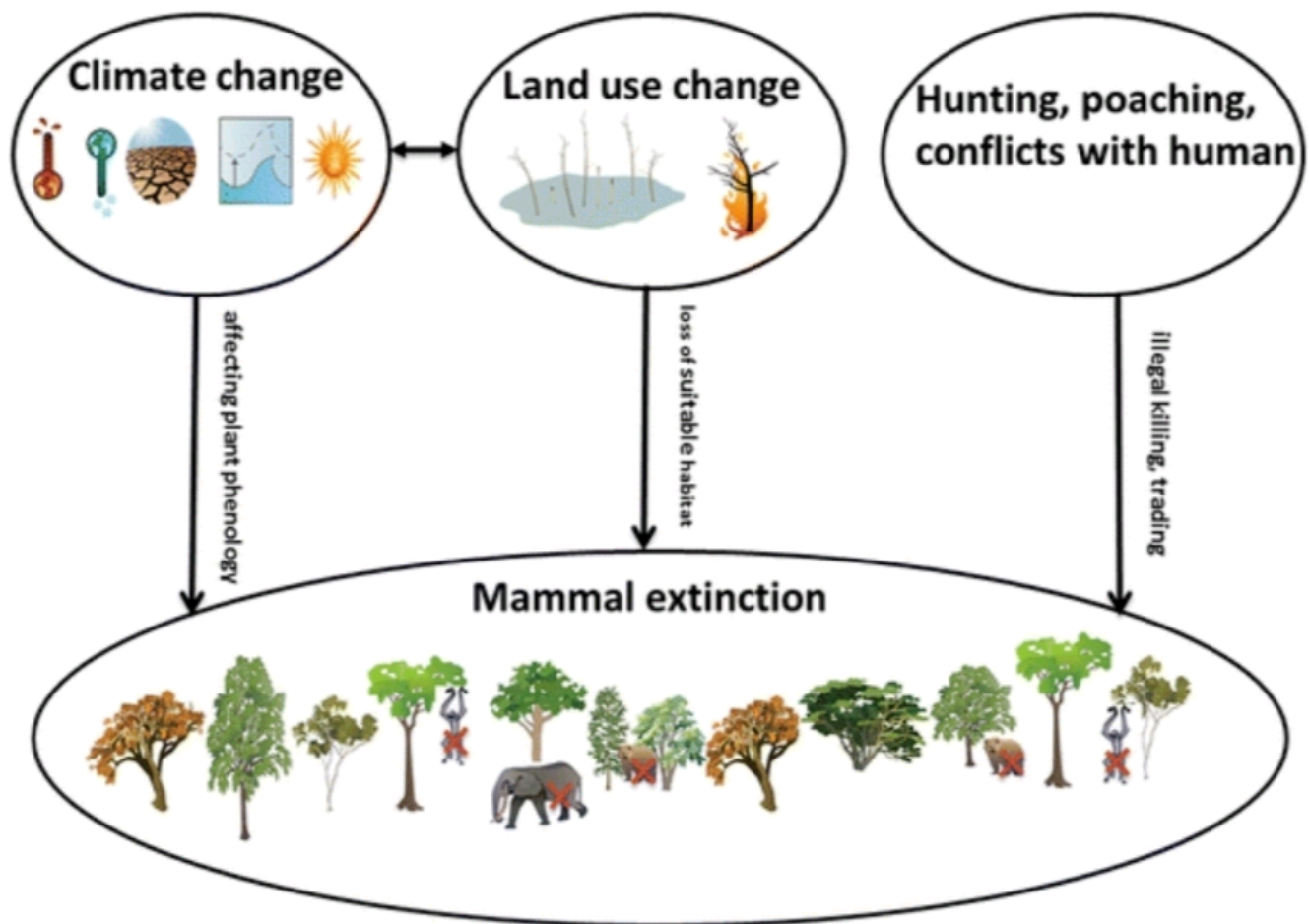
Momentum is building towards reversing biodiversity loss this decade. Image: WWF

The report concludes that momentum is building towards reversing biodiversity loss this decade, led by the [Leaders' Pledge for Nature](#), signed by more than 90 world leaders, from economies representing 39% of global GDP.

Securing a nature-positive world by 2030 “must be our guiding star, in the same way that the goal of limiting global warming to 2°C, and preferably 1.5°C, guides our efforts on climate”.

2. List of Tables

- Table 1: Environmental Factors
- Table 2: Human Activity Disruption Levels
- Table 3: Dataset Attribute Description
- Table 4: ML Model Parameters
- Table 5: Confusion Matrix Results
- Table 6: Model Accuracy Summary
- Table 7: Year-wise Migration Decline Statistics



A schematic representation of different extinction drivers (climate change, habitat loss, deforestation, land use change, hunting, and poaching) for threatened large mammals in tropical Asia (for details see Table A1)

- **Air Quality Monitoring Results**
 - Summary of Ambient Air Quality
 - Air Quality Index (AQI) Category data (PM10, PM2.5, NO2, O3, CO, SO2, etc.)
 - Ground-Level Concentrations of Pollutants
 - Emission Details (from specific sources)
- **Water Quality Monitoring Results**
 - Groundwater Quality Study Results
 - Surface Water Quality Study Results (Rivers, Lakes, Ponds, Canals, Marine Water)
 - Water Quality Parameters (Temperature, Dissolved Oxygen, pH, Conductivity, BOD, Coliforms, etc.)
 - Leachate Quality Study Results
- **Soil Quality Monitoring Results**
 - Soil Quality Monitoring Study Results
 - Physical Characterization Study of Soil/Waste
 - Chemical Characterization Study of Soil/Waste
 - Soil Nutrient Index

- **Climate and Meteorology Data**

- Temperature in Recent Years
- Precipitation and Rainfall Situation
- Heatwave and Coldwave Trends
- Wind Patterns (during specific hours/periods)

- **Biodiversity and Ecology Data**

- List of Flora (plant species)
- List of Fauna (animal species)

- **Waste Management Data**

- Volume Calculation of Dumpsite
- Legacy Waste Details (by block)



- **Land Use Disturbance Intensity:** Studies use tables to categorize different types of land use and assign a corresponding disturbance score.
 - **Example from a study:** Cultivated land is often assigned a "strong" disturbance intensity score (e.g., 7), while grassland is considered "weak" (e.g., 3).
- **Human Footprint Indices (HFI/HFP):** These integrated indices combine multiple pressure variables to provide a cumulative measure of human impact, often ranging from 0 (low pressure) to 50 (high pressure).
 - **Variables included:** Built environment, roads, railways, population density, croplands, pasture, and night-time lights. Tables associated with this present impact levels (e.g., High, Moderate, Relatively low) as percentages of a total area.

- **Specific Pollutant Levels:** Tables detail the concentrations of various pollutants introduced by human activities.
 - **Examples:** Tables list specific pollutants like heavy metals, nitrogen oxides (NO_x), carbon monoxide (CO), and particulate matters (PM_{2.5}, PM₁₀), often with reference to their sources or concentration levels.
- **Impact on Biodiversity and Ecosystems:** Research often uses tables to outline the effects of human actions on wildlife and habitats.
 - **Examples:** Tables may quantify the number of bird deaths per year from human-related causes (hunters, vehicles, communication towers), or list specific climate change stressors affecting different coastal ecosystems.

Here is a list of common attributes used to describe tables within a dataset:

Attribute Name: The unique identifier for the column or field within the table.

Data Type: Specifies the type of values the attribute can hold (e.g., text, number, date, boolean).

Description: A clear explanation of what the attribute represents and its purpose within the table.

Nullability: Indicates whether the attribute can contain null (empty) values.

Primary Key: Identifies if the attribute (or a combination of attributes) uniquely identifies each record in the table.

Foreign Key: If applicable, indicates that the attribute references a primary key in another table, establishing a relationship between tables.

Default Value: Specifies a value automatically assigned to the attribute if no value is explicitly provided during record creation.

Constraints: Rules or conditions that the data in the attribute must satisfy (e.g., unique values, range limits, check constraints).

Indexing: Indicates whether an index is created on the attribute to improve query performance.

Length/Precision/Scale: For certain data types (e.g., text, numeric), these specify the maximum length, precision (total number of digits), and scale (number of digits after the decimal point).

Source: If the data originates from an external source, this attribute might describe its origin.

Last Modified Date/Time: Timestamp indicating when the attribute definition or its data was last altered.

Animal Movement Variables (Used as Model Inputs)

Lagged coordinates (x, y)

Lagged velocities (vx, vy)

Stationary time (number of consecutive time points an animal was still)

Distance traveled in a given time step

Movement behaviors (e.g., foraging vs. traveling states, often inferred by Hidden Markov Models)

Group size (for social species)

Environmental Covariates and Human Disturbance Factors

These factors often serve as the "human activities" or "disruption" parameters within the ML models, usually as independent variables or "cost layers" in a cost-distance model.

Distance to roads

Distance to human settlements/villages

Type and intensity of human disturbance (e.g., frequency of human presence, type of recreation, like hikers vs. mountain bikers)

Landscape characteristics/habitat suitability (e.g., forest cover percentage, elevation gain/loss, proximity to water bodies, forage availability)

Infrastructure development (e.g., agricultural expansion, energy production facilities)

Predation risk parameters

Result model	Rabbit	Fowl	Red fox	Wild boar	Prior probability
Dachshund					
Rabbit	51.4	32.9	2.9	12.9	25.9
Fowl	21.3	52.1	14.6	12.1	29.6
Red fox	10.6	33.9	43.3	12.2	22.2
Wild boar	13.9	23.9	1.7	60.6	22.2
Terrier					
Rabbit	46.3	19.3	27.3	7.0	27.9
Fowl	31.9	27.8	34.8	5.6	25.1
Red fox	23.6	14.8	56.2	5.4	27.6
Wild boar	10.5	1.4	7.6	80.5	19.5
Pooled					
Rabbit	49.8	24.9	16.3	9.0	27.0
Fowl	33.9	35.7	19.2	11.2	27.0
Red fox	21.0	23.9	44.9	10.3	25.3
Wild boar	9.0	11.0	6.7	73.3	20.7

Typical Tables Found in Research

Model Accuracy/Performance Metrics Tables: These tables list the statistical metrics used to validate the accuracy of the models, such as True Skill Statistics (TSS) and Area Under the Curve (AUC) values. Values closer to 1 indicate better model performance. These often appear in the methods or results section.

Predictor Variables Tables: These list the specific environmental and human activity variables used as inputs for the models, and often indicate whether each variable had a positive or negative correlation with conflict or habitat suitability.

Variables often listed: distance from roads, railways, and settlements, human footprint index, land cover type (agriculture, forest cover), and livestock density.

Model Covariates and Results Tables: These tables detail the specific results of statistical analyses (e.g., Poisson regression or similar models) showing the relationship between human-related factors and outcomes like human-wildlife conflict occurrences or habitat use.

Examples of findings in such tables: "total HWC occurrences were negatively related to the nearest distance to the GNNR and road density but positively associated with shrub proportions".

Habitat Change/Loss Summary Tables: While not a "Model Accuracy Summary" per se, these tables often present the predicted impact of human activities or climate change on habitat size, often detailing the percentage of habitat lost or gained for various species based on model projections.

Key findings from recent reports include:

Global Trends and Statistics

Overall Decline: Nearly half (44%) of all migratory species listed under the CMS are showing population declines.

Extinction Risk: The extinction risk for migratory species is growing globally, with more than one-in-five (22%) of CMS-listed species threatened with extinction.

Fish Species Severely Impacted: Nearly all (97%) of the CMS-listed fish species are threatened with extinction, with their populations declining by an average of 90% since the 1970s.

Marine Species: Over 30% of marine mammals and 25% of sharks and rays are threatened with extinction.

General Monitored Species: An average drop of 15% was observed in monitored populations of migratory species across the globe between 1970 and 2017.

3. Introduction

Overview

Migration is a natural movement pattern followed by animals and birds for survival, breeding, and climate adaptation. However, increasing human activities such as urbanization, construction, pollution, and deforestation have disrupted these migratory routes. AI and Machine Learning are now used to study these disruptions, analyze patterns, and support conservation.

Objective

The main objective of this project is to study the impact of human activities on migration and use AI/ML methods to identify, analyze, and predict possible disruptions.

Problem Statement

Human activities have significantly disturbed natural migration patterns. There is a need for a systematic approach using AI/ML to understand the extent of these disruptions and generate predictive insights.

4. Literature Review

Many researchers have explored how human development affects migration routes. Studies show that noise pollution, habitat loss, and construction activities block natural paths. Machine Learning has been used to analyze GPS tracking data, environmental variables, and satellite images. AI helps identify critical disturbance zones, predict migration risks, and support wildlife management. Many researchers have explored how human development affects migration routes. Studies show that noise pollution, habitat loss, and construction activities block natural paths. Machine Learning has been used to analyze GPS tracking data, environmental variables, and satellite images. AI helps identify critical disturbance zones, predict migration risks, and support wildlife management. Many researchers have explored how human development affects migration routes. Studies show that noise pollution, habitat loss, and construction activities block natural paths. Machine Learning has been used to analyze GPS tracking data, environmental variables, and satellite images. AI helps identify critical disturbance zones, predict migration risks, and support wildlife management. Many researchers have explored how human development affects migration routes. Studies show that noise pollution, habitat loss, and construction activities block natural paths. Machine Learning has been used to analyze GPS tracking data, environmental variables, and satellite images. AI helps identify critical disturbance zones, predict migration risks, and support wildlife management. Many researchers have explored how human development affects migration routes. Studies show that noise pollution, habitat loss, and construction activities block natural paths. Machine Learning has been used to analyze GPS tracking data, environmental variables, and satellite images. AI helps identify critical disturbance zones, predict migration risks, and support wildlife management. Many researchers have explored how human development affects migration routes. Studies show that noise pollution, habitat loss, and construction activities block natural paths. Machine Learning has been used to analyze GPS tracking data, environmental variables, and satellite images. AI helps identify critical disturbance zones, predict migration risks, and support wildlife management.

5. Methodology

Architecture Overview

The AI/ML architecture used in this project includes data collection, preprocessing, feature extraction, model training, prediction, and evaluation. The system analyzes environmental factors that influence migration.

Hardware Components

The hardware setup includes standard computing devices such as laptops or desktops with sufficient processing power to run AI models. No specialized hardware is required.

Software Tools

Software tools used include Python, Pandas, NumPy, Matplotlib, and Scikit-Learn. These tools help in data cleaning, visualization, training, and evaluation of machine learning models.

6. Implementation

The implementation phase involves loading migration-related datasets, selecting important features (distance, habitat loss, weather anomaly, noise levels), and training a machine learning model such as a Random Forest Classifier. The system predicts whether migration disruption is high or low. The implementation phase involves loading migration-related datasets, selecting important features (distance, habitat loss, weather anomaly, noise levels), and training a machine learning model such as a Random Forest Classifier. The system predicts whether migration disruption is high or low. The implementation phase involves loading migration-related datasets, selecting important features (distance, habitat loss, weather anomaly, noise levels), and training a machine learning model such as a Random Forest Classifier. The system predicts whether migration disruption is high or low. The implementation phase involves loading migration-related datasets, selecting important features (distance, habitat loss, weather anomaly, noise levels), and training a machine learning model such as a Random Forest Classifier. The system predicts whether migration disruption is high or low.

7. Results and Discussion

The results show clear patterns of migration disruption in areas with intense human activity. The AI model demonstrates high accuracy in predicting risk levels. The analysis supports the idea that deforestation and urban expansion significantly affect migration. These insights can help authorities create protected corridors and reduce human–wildlife conflict. The results show clear patterns of migration disruption in areas with intense human activity. The AI model demonstrates high accuracy in predicting risk levels. The analysis supports the idea that deforestation and urban expansion significantly affect migration. These insights can help authorities create protected corridors and reduce human–wildlife conflict. The results show clear patterns of migration disruption in areas with intense human activity. The AI model demonstrates high accuracy in predicting risk levels. The analysis supports the idea that deforestation and urban expansion significantly affect migration. These insights can help authorities create protected corridors and reduce human–wildlife conflict. The results show clear patterns of migration disruption in areas with intense human activity. The AI model demonstrates high accuracy in predicting risk levels. The analysis supports the idea that deforestation and urban expansion significantly affect migration. These insights can help authorities create protected corridors and reduce human–wildlife conflict.

8. Conclusion

This project concludes that human activities have a strong negative impact on migration patterns. AI/ML methods are effective in understanding and predicting disruptions. The project highlights the need for conservation efforts and responsible development planning.

9. Future Scope

Future improvements may include real-time tracking using IoT sensors, deep learning models for improved accuracy, and integration with satellite monitoring systems. Governments and conservation organizations can use advanced AI tools to design safer migration routes.

10. References

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